CBSE Class 10 Mathematics — Areas Related to Circles

Time Allowed: 3 Hours **Maximum Marks: 80**

General Instructions:

1. This question paper contains $_{questions. All questions are compulsory.}$

1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. If the circumference of a circle is 176 cm, then its radius is:

(A) (A) 14 cm
(B) (B) 21 cm
(C) (C) 28 cm
(D) (D) 35 cm

2. The area of a circle whose diameter is 14 cm is:

(A) (A) 77 cm^2

(A) (B) 154 cm^2

(A) (C) 308 cm^2

(A) (D) 44 cm^2

3. The area of a quadrant of a circle with radius r is:

(A) $(A)2\pi r$
(B) $(B)\pi r^2$
(C) $(C)\frac{1}{2}\pi r^2$
(D) $(D)\frac{1}{4}\pi r^2$

4. If the perimeter of a semi-circular protractor is 36 cm, its diameter is:

(A) (A) 14 cm
(B) (B) 7 cm
(C) (C) 21 cm
(D) (D) 28 cm

5. The area of a sector of a circle with radius 6 cm and central angle 60° is:

(A) $(A)3\pi cm^2$

(B) $(B)4\pi cm^2$

(C) $(C)6\pi cm^2$

(D) $(D)12\pi cm^2$

6. Assertion (A): The area of a circle with radius 7 cm is 154 cm$^2.Reason(R):Theareaofacircleisgivenbytheformula\pi r^2$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

7. Assertion (A): The circumference of a circle is 44 cm, then its radius is 7 cm. Reason (R): Circumference of a circle is $2\pi r$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

8. Assertion (A): The area of a semicircle of radius r is $\frac{1}{2}\pi r^2$. Reason (R): A semicircle is half of a circle.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

9. Assertion (A): If the perimeter of a circle is equal to that of a square, then the area of the circle is greater than the area of the square. Reason (R): The area of a circle with a given perimeter is always the maximum possible area for any plane figure.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

10. Assertion (A): The length of an arc of a sector of a circle with radius r and angle θ is $\frac{\theta}{360} \times 2\pi r$. Reason (R): The circumference of a circle is $2\pi r$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

1. Find the area of a sector of a circle with radius 6 cm if the angle of the sector is $60°$. (Use $\pi = \frac{22}{7}$)

2. The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes.

3. Find the area of a quadrant of a circle whose circumference is 44 cm. (Use $\pi = \frac{22}{7}$)

4. If the sum of the areas of two circles with radii R_1 and R_2 is equal to the area of a circle of radius R, prove that $R_1^2 + R_2^2 = R^2$.

5. A chord of a circle of radius 10 cm subtends a right angle at the center. Find the area of the corresponding minor segment. (Use $\pi = 3.14$)

Section D: Long Answer (4-5 marks each)

1. Find the area of the shaded region in a circle of radius 14 cm, where a sector subtends an angle of $60°$ at the center. (Use $\pi = \frac{22}{7}$)

2. The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes.

3. Find the area of a quadrant of a circle whose circumference is 44 cm.

4. A chord of a circle of radius 10 cm subtends a right angle at the center. Find the area of the corresponding minor segment. (Use $\pi = 3.14$)

5. The radii of two circles are 8 cm and 6 cm respectively. Find the radius of the circle having area equal to the sum of the areas of the two circles.

6. A chord AB of a circle of radius 15 cm subtends an angle of $60°$ at the centre. Find the areas of the corresponding minor and major segments of the circle. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$)

7. An umbrella has 8 ribs which are equally spaced. Assuming the umbrella to be a flat circle of radius 45 cm, find the area between the two consecutive ribs of the umbrella.

8. Find the area of the shaded region in a square $ABCD$ of side 14 cm, where four circles are drawn with centers at the four vertices of the square, each having a radius of 7 cm.

9. A round table cover has six equal designs as shown in the figure. If the radius of the cover is 28 cm, find the cost of making the designs at the rate of Rs. 0.35 per cm^2. (Use $\sqrt{3} = 1.7$)